

Formula

Use the known σ to build the CI:

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

Example

Battery sample: $\bar{x} = 9.6$, $\sigma = 1.2$, $n = 36$, 95% CI $\rightarrow 9.6 \pm$

$$1.960 \cdot 1.2 / \sqrt{36} = 9.6 \pm 0.392 = (9.208, 9.992).$$

Interpretation

“We are 95% confident that the true mean battery life lies

between 9.21 and 9.99 hours.” If μ_0 falls outside this

interval, the two-tailed z-test at $\alpha = 0.05$ would reject H_0 .

Tips

- Use one-sided z critical values if reporting one-sided

bounds.

- Mention σ is known when citing the interval.